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Building Hybrid Pharmacometric-Machine Learning Models in Oncology Drug Development: Current State and Recommendations

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ABSTRACT

Classic and hybrid pharmacometric-machine learning models (hPMxML) are gaining momentum for applications in clinical drug development and precision medicine, especially within the oncology therapeutic area. However, standardized workflows are needed to ensure transparency, rigor, and effective communication for broader adoption. In this tutorial, we review pharmacometric (PMx) and machine learning (ML) reporting standards and evaluate them against hPMxML works in oncology contexts as a motivational example to identify current deficiencies and propose mitigation strategies for future efforts. The revealed gaps include insufficient benchmarking, absence of error propagation, feature stability assessments, and ablation studies, limited focus on external validation and final parametrization, and discrepancies between the performance metrics chosen and the original clinical questions. To address these, we propose a checklist for hPMxML model development and reporting, consisting of steps for estimand definition, data curation, covariate selection, hyperparameter tuning, convergence assessment, model explainability, diagnostics, uncertainty quantification, validation and verification with sensitivity analyses. This standardized approach is expected to enhance the reliability and reproducibility of hPMxML outputs, enabling their confident application in oncology clinical drug development, while fostering trust among all stakeholders.

1 | Introduction

The integration of artificial intelligence (AI) and machine learning (ML) into the pharmaceutical industry is positioned to transform drug development, offering new tools to enhance insights and improve clinical decision-making. Clinical pharmacology, for instance, might benefit from these methods to improve the adaptive design of Phase I trials [1], tailor product-specific guidance [2], or automatize non-compartmental analysis [3]. Health authorities (HAs) might see opportunities to establish a new regulatory framework [4] with AI/ML assisting regulatory decisions, as it occurred for the SAVE-MORE trial in COVID-19 patients [5] sponsored by the Center for Drug Evaluation and Research (CDER). Pharmacometrics (PMx), on the other hand, might profit from boosting its traditional inferential strengths with the predictive power of ML to enhance clinical outcome assessment [6–9]. In this sense, on top of classic AI/ML applications, PMx is experiencing a

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wave of methodological hybridization with AI/ML techniques as emphasized in [7, 10–16] and in software domain activities, such as [15, 17, 18]. Although developed for different scopes or contexts of use (CoU), the majority of current hybrid PMx-ML clinical applications can be broadly classified into three main categories, based on their goals of integrating PMx and AI/ML. The first category includes ML algorithms employed to scan extensive covariate pools against individual parameter estimates of PMx models (as targets). The process is intended to identify clinically meaningful and/or statistically influential covariates for subsequent inclusion in the PMx framework, effectively supplementing or accelerating traditional (bias-prone) covariate search methods such as stepwise covariate modeling (SCM) [11, 19]. The second category deals with AI/ML architectures trained on PMx model-derived metrics (AUC, Cmax, growth rates) and observed baseline or time-varying covariates to support exposure-response, regression (biomarker or pharmacokinetics-pharmacodynamics (PK-PD) metrics), survival analysis, or classification tasks [8, 12, 20–22]. The last category leverages AI/ML-based models to infer unknown dynamical equations from data for PK-PD and disease progression modeling, with the possible incorporation of inter-subject variability and the usage of new (undiscovered) high-dimensional modalities [7, 10, 23–25].

Building on these premises, cross-industry discussion groups (such as the IQ Clinical Pharmacology Leadership Group's AI/ML Working Group) and regulatory agency committees have been established to foster the integration of AI/ML principles in clinical development. Although, the aim of these initiatives is to promote alignment on AI/ML-based clinical products in high-risk CoU, most of their work is primarily focused on review of use cases of AI/ML [26–28], sidelining methodological must-haves or solutions to incorporate into the statistical analysis plan. The TRIPOD-AI [29], CONSORT-AI [30], and SPIRIT-AI [31] reporting guidelines and the PROBAST-AI form [32] are to date the most advanced documents discussing AI/ML model building from a technical viewpoint, with Food and Drug Administration (FDA)'s [33, 34] and European Medicines Agency (EMA)'s [35] primers bringing regulatory insights on top. FDA, in particular, has provided a list of general questions-to-be-addressed—on (1) human-led governance, accountability, and transparency; (2) data quality, reliability, and representativeness; (3) model development, performance, monitoring, and validation—to minimize pushbacks when AI/ML interventions are part of the submission package [33]. In this sense, adoption of a risk-based credibility framework based on a "model influence-versus-decision consequences" matrix is strongly encouraged at each developmental step to meet these regulatory expectations [34]. Nonetheless, all these "good practice" efforts tend to be quite universal and unspecific to accommodate AI/ML models in a broad pharmaceutical context, from clinical research to manufacturing, and usage., from classical regression tasks to generative AI. Thereby, they may focus less on the specificities of model-informed drug development (MIDD) field and related data preparation workstreams, or on *ad hoc* precautions necessary when AI/ML is combined with other modeling techniques, often used in PMx. In this sense, hPMxML model practitioners are currently working in isolation, approaching the hybrid modeling in a plethora of ways, due to the lack of unified reporting and developmental practices specific to these new methods, for example, error propagation handling and causality. Because of this setup, the full potential of this modeling partnership might be hindered or slowed down as transparency, reproducibility, and clinical adoptability are negatively impacted.

Building on this gap, the present article is intended to advocate a mindset shift in the current ML and PMx landscape in favor of workflow standardization and harmonization, as Figure 1 shows. In particular, our aims are fourfold. First, to equip pharmacometricians willing to incorporate ML techniques in some daily tasks, as well as ML experts willing to apply their know-how to clinical pharmacology, with a roadmap for hPMxML model building and validation steps that ensure compliance with Good Clinical Practice (GCP) standards and regulatory expectations. Second, as collaboratively crafted by PMx and AI/ML experts, to align PMx and ML communities in terms of terminology and respective workflows, thereby easing cross-functional work within different divisions of the pharmaceutical industry. Third, to avoid a "lost-in-translation" case from fragmented PMx-ML exercises to clinical development routine [36], guaranteeing that questions for real-world usage are addressed technically and clinically. Fourth, in line with cross-setting expert opinions on the PMx-modelers-of-the-future [37], to provide educational material for training the next generation of ML-aware pharmacometricians. While our work stems from an analysis of the current status of hPMxML in oncology as a motivational case, the proposal is intended to be guidance for scientists across various disease domains.

2 | Review Workflow for PMx and ML Guidelines

Well-established technical guidelines and working protocols were reviewed to identify must-haves and nice-to-haves reporting elements for the PMx and AI/ML communities, respectively.

For the AI/ML domain, insights on optimal development and reporting standards were derived from Collins et al. (TRIPOD-AI) [29], Norgeot et al. (MI-CLAIM) [38], Luo et al. [39], and Navarro et al. [40, 41]. For specifics on prognostic studies, Dhiman et al. [42] and Altman et al. [43] were considered, while the PROBAST-AI document [32] was utilized to inspect the ML landscape in terms of biases and mitigation strategies.

Musuamba et al. [44], Lim et al. [45], Schmidt et al. [46], and Bergsma et al. [47] were reviewed to gain a pictorial overview of established PMx workflows, from model assumptions to model credibility assessment and validation. Reporting standards for traditional statistical models employed in prognostic studies were derived from Kempf et al. [48] and Janssens et al. (GRIPS) [49], while expectations on the selection of validation metrics came from Steyerberg et al. [50].

To facilitate meaningful cross-domain comparisons between pharmacometric and machine learning approaches, the identified documents were organized according to a stepwise modeling

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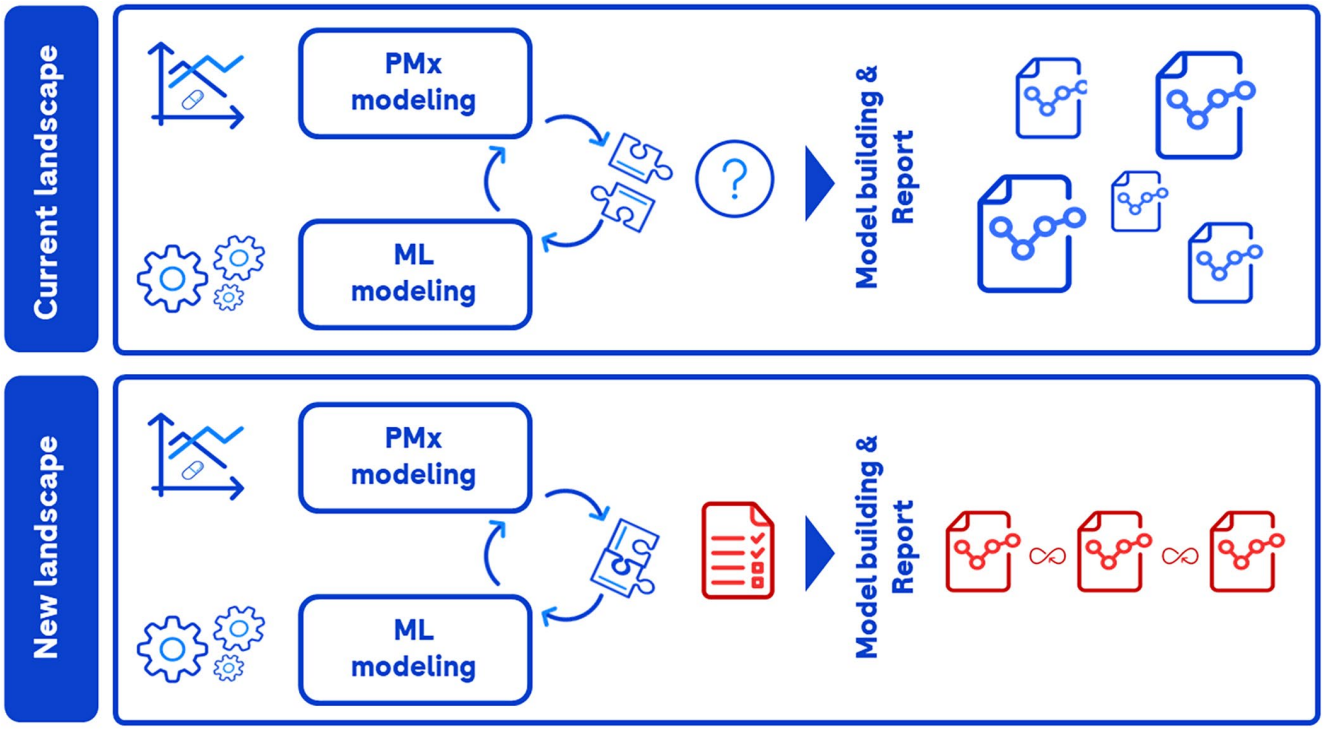


FIGURE 1 | The vignette compares the current approach in integrating PMx and machine learning modeling with our envisioned one. While today's efforts are isolated with modelers producing their model report and analysis at the cost of generalizability, reproducibility, and communicability, the introduction of a checklist of good development and validation practices may help in harmonizing results, while endorsing this new hybrid modeling in high-risk contexts.

framework (see keyword paragraph and Tables S1 and S2). This framework reflects the typical progression of model development, including: defining the question of interest or context of use, data selection and preparation, specification of features or covariates, model development and training, evaluation and validation, performance assessment, and analysis of uncertainty, sensitivity, and model limitations. These stages were aligned with concepts outlined in the FDA draft guidance on Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products [34]. An occurrence analysis was performed to identify commonalities and distinctions across the two domains. Elements found exclusively in one domain and not addressed in the guidance were treated as domain-specific in the comparison.

3 | Review Workflow for hPMxML Model Candidates and Evaluation Criteria

Publicly available hPMxML models in oncology clinical development contexts were assessed for reporting completeness and quality against the unified PMx and ML standards previously identified. An initial set of candidate works was identified by the four sequential queries (i) "(Machine Learning OR Machine Learning-based OR Interpretable Machine Learning OR Explainable Machine Learning OR Neural Differential Ordinary Equations OR NODE OR Neural-ODE) AND (Pharmacometrics)", (ii) "(Pharmacometrics) AND (Machine Learning) AND (Cancer OR Tumor OR Oncology)", (iii) "(Neural ODE) AND (Cancer OR Tumor OR Oncology)", and (iv) "(Survival Prediction) AND (Machine Learning) AND (Clinical Pharmacology) AND (Oncology) NOT (Genetics AND Radiomics AND Metabolomics AND Imaging)". The queries were restricted to a period ranging between 01 January 2020 and 15 December 2024, with evident applications to the oncology disease area. Reviews, comments, editorials, tutorials, comparative/observational studies, case reports were discarded, as well as original articles not presenting a clear quantitative use case nor a clinical purpose nor cancer-specific synthetic data. Candidates making use of bioinformatic processing of omics/genetic sequencing or imaging techniques were deemed not eligible. Examples of unsupervised AI/ML were considered beyond the intention of this work and therefore dropped. For a full description of the review implementation, refer to the PRISMA flow diagram in the Supporting Information (Figure S1).

Each high-level element resulting from the comparison of PMx and AI/ML workstrams was dissected to identify objective metrics for assessing the selected models. Evaluation metrics included reporting of data description (with delineation of applied data manipulations), rationale for performance metric selection, presence of an uncertainty quantification intended as a confidence interval for results, statements on validation techniques and dataset representativeness, standards of model explainability and interpretability, presence of model verification via sensitivity or robustness analysis, benchmarking, and code sharing. Ad hoc metrics emerging from the integration of PMx with AI/ML, that is, report of causal findings and rationale for error propagation handling, were also added. Categories scoring less than 50% in a normalized frequency analysis of report completeness across identified published works were considered inadequately conformed.

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TABLE 1 | Development and reporting pillars (rows) from PMx versus AI/ML (columns) best practices. A cross means that the descriptor is lacking, while a tick means that it is consistently present in all evaluated domain-specific documents. A yellow cell highlights a descriptor common to both domains but with a slightly different meaning and action needed.

	PMx workflow	ML workflow
CoU-specific model development plan	✓	✓
Data selection for validation based on comparator credibility	✓	✓
Data augmentation	✗	✓
Minimum generalizability sign via data splitting	✗	✓
Model credibility generated from assumption declaration	✓	✓
Optimization algorithm termination as part of model selection	✓	✓ Might be discriminative for some architectures only, such as neural network
Reference to a reference model to evaluate the gain	✓ Incrementally while model development	✓ In parallel across architectures with different complexity
Uncertainty quantification and management	✓ In the form of bootstrap or repeated covariance matrix sampling	✓ In the form of bootstrap, variance-, ensemble-based methods or conformal prediction
Model performance criteria based on explainability	✓ In the form of unexplained inter-individual variability	✓ In the form of functional form unraveling or final covariate plausibility
External validation	✓ Performed at different time or for CoU (for instance, via Bayesian feedback), but not required to endorse the model	✓
Benchmarking	✗	✓
Statistical adequacy of code and verification	✓	✓
Code and model parameter disclosure	✓	✓

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4 | Identified Similarities and Differences in PMx and ML Workstreams

Routinely adopted checklists for optimal development and validation of PMx and ML models emphasized substantial similarities in building precautions and reporting standards, alongside core specificities. Grounding on both techniques, we argue that hPMxML practitioners should be aware of these distinctions, either methodologically and terminology-wise, before commencing any modeling endeavor and should be compliant to both while designing the analysis plan. In this sense, as Table 1 shows, PMx and AI/ML workflows agree on (i) having a CoU-specific plan, (ii) declaring assumptions towards model credibility criteria, (iii) justifying the statistical adequacy of the framework and its verification via sensitivity analysis and robustness check, (iv) demonstrating the cases where the model fails (model boundaries), and (v) disclosing code and model parameters. While the alignment also extends to the quantification of any uncertainties, this aspect might be handled differently by each domain. With the bootstrap technique bilaterally adopted, PMx has found alternatives in Markov Chain Monte Carlo (MCMC) sampling, Sampling Importance Resampling (SIR) methods [51], asymptotic distribution properties, and/or clinical trial simulation plots generated from covariance-based samplings, while AI/ML in variance/ensemble-based methods [52] or conformal prediction [53] depending on the model architecture. Besides, although both PMx [54, 55] and AI/ML guidelines theoretically advocate for external validation to ensure model generalizability, in practice, this step is much more established in the AI/ML field. Indeed, even when PMx models are intended for predictive tasks, validation against other datasets might be postponed or applied to different CoU [56], with in-distribution validation in the form of (prediction-corrected) Visual Predictive Check (pc-VPC) [57], Numerical Predictive Check (NPC) or Posterior Predictive Check (PPC) [58] implemented as a faster supplement. Moreover, while AI/ML routinely performs data splitting for internal validation purposes, PMx often uses the entire dataset (hence, no data splitting) for model development, particularly in early phases, to avoid any harmful loss of data that might impact covariate detection and estimate precision. Besides, AI/ML protocols put emphasis on benchmarking the product outputs against competitors and on unraveling a contextual model explainability, which are usually never sustained or taken for granted in PMx, respectively. Finally, while data augmentation via oversampling strategies is commonly applied in AI/ML to mitigate imbalances or enrich portions of interest in the data, PMx routine discourages any data modification of this type (apart from synthetic entries for data imputation) [54] to avoid biases on model parameter estimates.

5 | State of the Field for hPMxML Models in Oncology

To illustrate the current status of the hPMxML literature against discussed PMx and ML must-haves and quality checks, an extensive review of oncology hybrid models was carried out. 29 works (see Table S3) were deemed suitable to be included in our exemplificative analysis. For 2 (7%) of them, the primary objective was methodological, with test cases on in silico or real oncology data, while for the remaining, it was clinical. 14 (48%) included clear components from both AI/ML and PMx/statistical modeling, with the large majority using PMx-derived metrics as either inputs or targets for the AI/ML intervention and 2 implementing neural-ODE techniques to complete an ordinary differential equation (ODE) system. 18 (62%) papers had the primary AI/ML component used for regression, half of which were for time-to-event settings, and 11 for binary or multi-class classification.

A comprehensive list of metrics, derived from PMx and AI/ML workflows and expectations, was used to evaluate the reporting quality of the selected hPMxML within a drug development and regulatory framework. For each metric, we verified if reporting details were met or not. Figure 2 shows the results of our evaluation through barplots counting the frequency of comprehensively reported items across the analyzed models. Major deficiencies (below the average cut-off of 50%) were found in error propagation handling, robustness assessment (such as feature stability or multi-start consistency), metric completeness, benchmarking, and material sharing. While we found the clinical question clearly stated in the paper introduction for all cases, we recognized a general lack of compliance to the ICH E9(R1) addendum (estimand definition in the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use) [59], especially for the declaration of the population-level summary measure and intercurrent event handling strategy when applicable. We acknowledged a comprehensive data description in terms of the clinical context (trial or treatment details, population characteristics, predictor and clinical endpoint rationale), but a milder one in terms of possible data manipulation techniques applied (average ~75%). We found internal validation, whether in the form of a training/test split or of cross-validation techniques (most employed) or of jackknife validation, fully covered, but less than 25% incidence of external validation. Comparison of AI/ML architectures was achieved in roughly 55% of works, with 7 manuscripts showing benchmark efforts against competitor models or SoC techniques (e.g., Stein pharmacodynamic model in oncology or SCM in covariate searching). The majority of classification and survival analysis works missed providing a strong rationale for the selection of the predictive performance metric (area under the Receiver Operating Characteristics curve (AUCROC) vs. Precision-recall curve in imbalance set, discrimination (AUC/Cindex) vs. calibration metrics), while alignment between research questions and metrics was fully achieved in regression tasks. Comparison with literature expectation or upfront definition of successful criteria as per CoU were almost never discussed. Model explainability was well analyzed (> 75%), with the latest technologies, such as SHapley Additive exPlanations (SHAP) values [60], which allow for either local interpretability and causal insights via conditional effects on a feature on the selected outcome, still under-used (< 18%). In terms of model inspections, we found that sensitivity analysis on thresholds or cutoffs was quite well performed (almost 75% occurrence), although rooms for possible user-based biases in model development could still be present, while subgroup analysis was approached mostly via subgroup-specific architectures only or ML-derived stratification. No analysis was conducted on understanding which cohorts were better predicted by a unique model. Error propagation was never discussed, leading to underestimation of the uncertainty and inability to decouple the source of variability. Finally, no works

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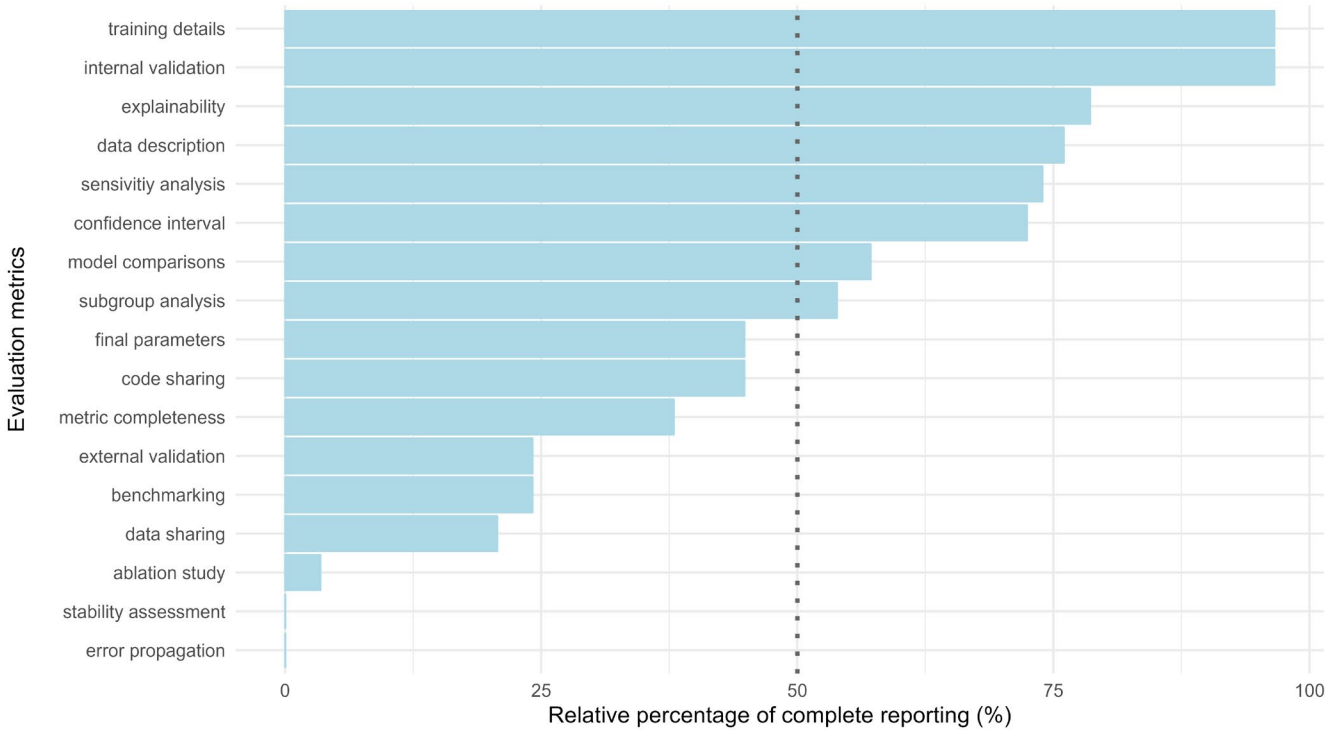


FIGURE 2 | The barplot shows the percentage of hPMxML models presenting reporting standards per topic as expected from the reviewed PMx and AI/ML guidelines. Percentages are normalized on the paper subset for which they might be applicable.

attempted a robustness assessment via a multi-start approach nor feature stability checks.

6 | Checklist Proposal for hPMxML Model Building

Despite the advantages and interest in hPMxML methodologies in clinical development contexts, the need for harmonization is still unmet. To address this demand and to guarantee the trustworthiness and credibility of these methods, we present a comprehensive checklist for hPMxML model building (Table 2), primarily intended to accommodate randomized clinical trial (RCT) datasets as the backbone of data-driven drug development. The proposed checklist is structured in seven blocks—clinical setup, data preprocessing, model training, model validation, secondary analysis, applicability domain, and benchmark studies—each including distinct and detailed subpoints. Descriptors were sign-coded (**Δ** = mandatory, **O** = optional) based on their regulatory weight—as derived from [34]—to alert users on which tasks to prioritize. This is expected to be particularly beneficial in cases where development time is tight or when users first approach this new hybrid modeling modality and might feel overwhelmed by the number of listed items.

7 | Main Considerations

An upfront deep knowledge of the CoU is a must-have condition to develop ML-based models as it drives most of the preliminary assumptions, technical choices, and benefit–risk evaluations along the full model development process. In this sense, we encourage the frame of the clinical question—target population, endpoint of interest, treatment conditions, summary measures, and intercurrent event [59]—and the definition of "model success" criteria to be jointly agreed a priori by clinical stakeholders and modelers, based on risk appetite and competitor performances. In addition, once the intended model usage is clearly defined, we suggest a critical and statistical analysis of the final filtered database to validate model utility toward goal achievement. In this sense, we refer to [71, 72] for simulation studies on minimum sample size in clinical AI/ML settings and to [73] for R-based computational support in the calculation. In addition, to gain a preliminary understanding of the overfitting likelihood for the planned setup, we refer to [74] for an overview on covariate pool versus sample size in specific classification contexts. In terms of communicability and usability awareness, a clear discussion on the selected sample size and secondary analysis on performance degradation when the number of training examples is reduced is nonetheless beneficial to provide indications on which drug development stages the model could serve.

To ensure reproducibility, we recommend that data engineering processes be documented as a sequential list of applied transformations. In instances where data manipulation is not conducted prior to AI/ML training, an explicit statement should be included to inform practitioners and prevent any misconceptions. In this regard, while data imputation is typically performed before model development, its implementation may vary by model type. For example, NONMEM scripts and tree-based methods handle missing data automatically, whereas Lasso does not. A more transparent and unbiased—though time-consuming—approach would be to conduct imputation separately for the development and (cross-)validation sets/folds, integrating it as part of model training.

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TABLE 2 | The table summarizes the 7-pillar checklist we designed to facilitate a harmonized development and validation of hPMxML models. Mandatory items are marked with **Δ**, optional ones with **O**. The last three columns labeled as Cat correspond to the three categories of hPMxML models presented in the introduction, namely Cat. I = covariate screening ML methods having PMx-derived Empirical Bayes Estimates (EBEs) as targets, Cat. II = predictive or classification models with clinical and PMx-based regressors, and Cat. III = neural-ODE systems to compensate for missing mechanistic terms.

1. Clinical setup (exemplary reference: [61])	**Cat. I**	**Cat. II**	**Cat. III**
1.1 Intended usage			
• 1.1.1 Clinical questions to be addressed and their relevance in terms of MIDD and decision-making as defined by clinical stakeholders **Δ** (define the question of interest)	☐	☐	☐
• 1.1.2 A priori definition of the successful criteria for model usability/adoptability as jointly defined by clinical stakeholders and modelers in terms of the clinical phase and disease severity, with reference to gold standard target product profile in the same context of use (CoU) (if any) **Δ** (define the CoU)	☐	☐	☐
• 1.1.3 Indication whether the model will be developed retrospectively or prospectively, with inferential, predictive, or counterfactual purposes, within a supervised or unsupervised learning methodology, as following from the clinical questions **Δ** (describe the learning methodology)	☐	☐	☐
1.2 Estimand definition			
• 1.2.1 Motivation for the choice of the target population (severity, eligibility criteria, portfolio strategy) and its alignment with the intended clinical questions **Δ** (define the CoU)	☐	☐	☐
• 1.2.2 References to possible subgroups (genetic-, biomarker-based, ...) that might mechanistically behave in a different manner **Δ** (define the CoU)	☐	☐	☐
• 1.2.3 Explanation of both endpoint and quantity of interest (effect size) and the related interpretation (*example: we considered death from any cause as the primary endpoint for the analysis, which ensures unbiasedness and across-study comparability, but has limited specificity. Since our goal was to compare cross-arm efficacy, the assumption is appropriate*) **Δ** (define model inputs/outputs)	☐	☐	☐
• 1.2.4 *[If relevant]* explanation of the chosen censoring strategy and its implication on result interpretation (*example: we assumed no intercurrent events, thus a treatment policy, by setting to censored patients experiencing treatment switch/discontinuation/lost-of-follow-up during the study duration. While this maintains randomization, using ITT over actual treatment will not be reflective of the protocol and might introduce bias from informative censoring*) **O**	☐	☐	☐
• 1.2.5 *[If applicable]* rationale for the choice of the metrics for model interpretation as related to the intended usage, clinical interpretability, and standard-of-care in the same CoU **Δ** (describe the CoU)	☐	☐	☐
2. Data processing (exemplary reference: [62])	**Cat. I**	**Cat. II**	**Cat. III**
2.1 Pre-processing metrics			
• 2.1.1 Reporting of pooled and trial-stratified statistics and summarizes for all the data of interest (whether longitudinal measurements, covariates and clinical targets): mean + standard deviation (SD) or median + median absolute deviation (MAD)/interquartile range (IQR) and skewness level for continuous variables, level frequency for categorical variables. The statistics should be provided for either training and test sets, separately, and for any subgroups of interest **Δ** (describe how the development data is fit for the CoU)	☐	☐	☐
• 2.1.2 Reporting of pair-wise/multi-wise correlation analysis (disregarding potential missing data) with the rationale for the selection of the statistical test for continuous variables and the approach to inspect continuous-categorical dependency **O**	☐	☐	☐
• 2.1.3 Reporting of pooled and trial-stratified percentage of missing measurements with diagnostic on missing data patterns and rationale of their impact on model development (informative censoring) and result interpretation (*e.g., VIM [63], Amelia [64], mice [65] R packages have options for pattern visualization*) **Δ** (describe the data management practices)	☐	☐	☐

(Continues)

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TABLE 2 | (Continued)

2.2 Data filtering			
• 2.2.1 Rationale for dropping covariates or longitudinal observations, with possible clinical implications and comparison with previous strategies from the same CoU **Δ** (describe the data management practices)	☐	☐	☐
• 2.2.2 *[If applicable]* if dropping comes from statistical analysis and implies decisional thresholds, justification for the chosen threshold should be provided (*example: columns having more than X% of missing data were dropped as studies XYZ showed a deterioration of model development in setting with ≥ X% NAs and other full-entry covariates C1, ..., Cn were sufficiently correlated to act as surrogate*) **Δ** (describe the data management practices)	☐	☐	
2.3 Data engineering			
• 2.3.1 Reporting of the sequential order of the transformations applied to the covariates or longitudinal measurements **Δ** (describe the data management practices)	☐	☐	☐
• *2.3.2 [If applicable]* if data imputation has been applied, reporting of the adopted technique (single vs. multiple vs. partial imputation, maximum likelihood imputation, subgroup-based vs. full dataset-based imputation, domain knowledge-based vs. model-based imputation etc.) and of main deviations between pre- and post-imputation in terms of mean/median/mode and dispersion. For model-based techniques on complex datasets, such as real world data (RWD) sources, reporting of the implementation specificities of the underlying regression step. In cases where RWD are used to replace missing data on comparative effectiveness with RCTs or on longitudinal outcomes, reporting of imputation validity in terms of data similarities **Δ** (describe the data management practices)	☐	☐	
• 2.3.3 *[If applicable]* if skewness reduction has been applied, reporting of technical details, such as the transformation explored and the final one applied to each covariate, and percentage reduction in skewness compared to pre-transformation o	☐	☐	
• 2.3.4 *[If applicable]* if any scaling has been applied, reporting of necessary details to reproduce the transformations (e.g., final min and max value for each continuous covariate, mean and standard deviation for each continuous variable for normalization (MinMax scaling) and standardization, respectively, chosen reference level with clinical or statistical justification for dummification of categorical (binary/multi-level) covariates) o	☐	☐	☐
2.4 Post-process metrics			
• 2.4.1 Reporting of summary metrics for final (post-processing, post-cleaning) data, whether in terms of longitudinal observations or of input covariates and clinical target with protocol-based motivations in case of marked discrepancies in covariates and a statistical discussion of the extent of population-shift (violation of machine learning assumption) in case of target discrepancies. If a subgroup analysis is planned, reporting of each stratum final statistics as well **Δ** (describe how the development data is fit for the CoU)	☐	☐	☐
2.5 Reproducibility			
• 2.5.1 Reporting of software (package/library) version control and lineage as used to achieve data processing sub-items **Δ** (as per data package in regulatory submission)	☐	☐	☐
3. Model training (exemplary reference: [66])	**Cat. I**	**Cat. II**	**Cat. III**
3.1 Model definition			
• 3.1.1 Reporting of all candidate models (with references to previous usage in the same CoU) and statistical justifications for the selection followed from the exploratory data analysis (EDA) step **Δ** (describe model architecture)	☐	☐	
3.2 Threshold selection			
• 3.2.1 *[If applicable]* rationale for the selection of any threshold in the analysis plan (landmark, horizon, statistical cut-off), possibly with references to protocol design or statistical analysis plan (SAP) **Δ** (describe model parameters)	☐	☐	

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TABLE 2 | (Continued)

3.3 Model calibration			
• 3.3.1 *[If applicable]* reporting of splitting strategies or cross-validation techniques employed and relative technical details, such as rationale for the k-fold selection as related to statistical power, number of repetitions **Δ** (describe how the development sets were split)	☐	☐	
• 3.3.2 *[If applicable]* reporting of any weights (e.g., propensity score weighting) or resampling techniques (e.g., oversampling, undersampling, hybrid strategies) planned in the model training with statistical and clinical rationale for their introduction and function arguments used for their computation. Clear distinction between application of resampling techniques for learning imbalance or subgroup-imbalance (sex, race, co-morbidities, etc.) towards regulatory fairness audits should be provided (*example: considering the binary classification task the tool is intended for, the training set imbalance (ratio 1:4) is mitigated* via *an oversampling approach with a 25% increase of the minority class* (1) *equally sampled from sex- and age-stratified subgroups given the known clinical influence of sex and age [cit XYZ] on the indication management*) **Δ** (describe model parameters)	☐	☐	
• 3.3.3 *[If applicable]* reporting of the hyperparameter tuning details, such as employed technique, search space constraints, implementation within the cross-validation schema, and of the final best parameters as retrieved from the chosen metric **Δ** (describe training hyperparameters)	☐	☐	☐
• 3.3.4 *[If applicable]* reporting of the covariate selection details, such as wrapper versus embedded technique, additional metric-dependent arguments, and of the final selected covariates **Δ** (describe feature selection process)	☐	☐	
• 3.3.5 *[If applicable]* reporting of the selected loss function and of any regularization techniques acting on it, with justification for their introduction and technical details on the setup. If fairness bias is intended to be mitigated via optimization function adjustment, that is, via fairness-aware log losses such as counterfactual logit pairing, a clear statement should be provided **Δ** (describe any loss function used for model design, performances metrics used to evaluate the model, and techniques employed to prevent over- or under-fitting)	☐	☐	☐
3.4 Convergence assessment			
• 3.4.1 Reporting of the implementation details for model optimization step, such as rationale for the selection of the optimization algorithm and for the chosen setup for max-evaluation flag and eventual additional arguments **o**	☐	☐	☐
• 3.4.2 *[If applicable]* reporting of a metric-versus-iteration plot to confirm the adequate setup of the maximum number of evaluations and to demonstrate lack of metric fluctuations that might reflect suboptimal local minima **o**	☐	☐	☐
3.5 Model diagnostic			
• 3.5.1 Reporting of observed versus predicted values (e.g., *contingency table/calibration curves for binary tasks, correlation plots for regression tasks, ...*) **Δ** (describe the performances metrics used to evaluate the model and the agreement between the model prediction and the observed data)	☐	☐	☐
• 3.5.2 Reporting of stability analysis for covariate selection across cross-validation folds or bootstrap schema with a statistical and clinical rationale for the selected similarity metric to evaluate ranking/occurrence stability (*example: ordered list (OL) score was employed to quantify the similarity of selected covariates and it was preferred over correlation-aware methods as correlated clinical covariates like XYZ might be collected with different procedure, hence different costs and patient burden, thus not being similar in the clinical practice*) **o**	☐	☐	

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TABLE 2 | (Continued)

3.6 Uncertainty quantification			
• 3.6.1 Reporting of the technique employed for model confidence calibration, with clear distinction between predictive (epistemic) and estimation (aleatoric) uncertainty, the underlying rationale for the selection, and details on the confidence class generation/significance level setup/conformal metric/bootstrap samples. If this step is missed in a high-risk CoU, reporting of a rationale for not equipping the model with a confidence notion **Δ** (describe the process by which the uncertainty of model predictions was estimated)	☐	☐	☐
3.7 Reproducibility			
• 3.7.1 Reporting of software (package/library) version control and lineage as used to achieve model training sub-items **Δ** (as per data package in regulatory submission)	☐	☐	☐
4. Model validation (exemplary references: [20] for validation schema, [67] for explainability)	**Cat. I**	**Cat. II**	**Cat. III**
4.1 Validation metric			
• 4.1.1 Depending on the intended usage and the characteristics of the dataset (imbalance), reporting of the rationale for the validation metric selection (discrimination vs. calibration, recall vs. recall parity or false positive rate vs. false positive rate parity in case of fairness bias mitigation) with references to previous metrics employed in the same CoU, possible recommendations from HA guidelines, and clinical-practice interpretability. For "evaluating for bias" tasks on subgroups of clinical interest, clear reporting of the adopted mitigation strategy out of demographic parity, equality of opportunity, and counterfactual fairness (*example: a discriminative metric was selected for the tool predictive performances as the intended clinical usage is risk stratification, thus it is essential to evaluate model ability in discriminating between high-, moderate-, and low-risk subgroups in order to set up most suitable therapy*) **Δ** (describe the rationale for the chosen model evaluation metrics)	☐	☐	☐
• 4.1.2 Reporting of the predictive performances for each test-set **Δ** (describe the agreement between the model prediction and the observed data)	☐	☐	☐
• 4.1.3 *[If applicable]* reporting of the confidence class distribution for the test-set or the test-subset percentage passing the confidence analysis at the predefined significance level **Δ** (describe descriptors to quantify confidence or uncertainty)	☐	☐	☐
• 4.1.4 *[If applicable]* reporting of the confidence intervals for model predictions with reference to the approach used to collect output distribution (e.g., bootstrap) and to the significance level as rule-of-thumb in the same CoU **Δ** (describe descriptors to quantify confidence or uncertainty, i.e., performance metrics should be provided with confidence intervals)	☐	☐	☐
4.2 Model explainability			
• 4.2.1 Reporting of the overall/global covariate importance ranking with comparison to previous work in the same CoU/indication and with clinical interpretability where possible (clinically-unjustified or never-seen-before covariates deemed as important must be reported with discussion on a plausible dataset-dependency) **Δ** (describe feature selection process)	☐	☐	
• 4.2.2 Reporting of the covariate value-versus-outcome relationship with discussion on the clinical feasibility of the directionality **o**	☐	☐	
• 4.2.3 Reporting of any findings collected from the covariate importance at individual- and global-level with details on the methodology (metrics, packages) in place to derive the ranking (e.g., compensating/delayed/up-and-down effects of longitudinal feature importance for survival settings in survex R package [68]) **o**	☐	☐	
• 4.2.4 *[If applicable]* if a causal knowledge is available for key covariates and a direct acyclic graph can be drawn, reporting of any changes in the feature importance ranking occurred with the introduction of the causality concept (shapFlex R package [69]) **o**	☐	☐	

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TABLE 2 | (Continued)

• 4.2.5 Reporting of any explanation of the trained input–output relationship with implementation details. If symbolic regression is applied [70], reporting of a rationale for the selection of the functional family o	☐	☐	☐
4.3 Reproducibility			
• 4.3.1 Reporting of software (package/library) version control and lineage as used to achieve model validation sub-items **Δ** (as per data package in regulatory submission)	☐	☐	☐
5. Secondary analysis (exemplary references: [22] for subgroup analysis, [23] for sensitivity analysis)	**Cat. I**	**Cat. II**	**Cat. III**
5.1 Subgroup analysis			
• 5.1.1 Reporting of any deterioration/improvement seen when the model is fully tested on pre-specified subgroups with a rationale of the outcome as related to subgroup frequency in the development set **Δ** (describe the CoU and potential bias)	☐	☐	☐
5.2 Sensitivity analysis			
• 5.2.1 *[If applicable]* although a rationale for any threshold/cut-off is being justified above, reporting of any deterioration/improvement in the performances when different cutoffs are adopted, with a clear clinical rationale for the selection of the sensitivity analysis domain and a discussion of the likelihood/impact of these alternative scenarios o	☐	☐	
5.3 Ablation studies			
• 5.3.1 Reporting of any deterioration/improvement seen when key features or components are artificially dropped one-at-the-time or in combinations. If the investigation is carried out on a subset only, provide a rationale for the subset identification o	☐	☐	☐
6. Applicability domain (exemplary reference: [20])	**Cat. I**	**Cat. II**	**Cat. III**
• 6.1 Rationale for the test-set representativeness of the intended CoU with references to eligibility criteria-versus-clinical practice and external datasets and proofs of the data independence **Δ** (describe the applicability of the test data to the CoU)	☐	☐	☐
• 6.2 Reporting of any deterioration/improvement seen when the model is tested on instances that are distant from the reference ones (data drift), with statistical details on the outlier definition (metric used) and their clinical relevance **Δ** (describe the applicability of the test data to the CoU and potential bias)	☐	☐	☐
7. Benchmark studies (exemplary reference: [23])	**Cat. I**	**Cat. II**	**Cat. III**
• 7.1 *[If applicable]* reporting of performative metrics in the same CoU from previous works with details on their validation setting, model complexity, and training set size. Compare their performances with own ones, otherwise discuss the direct-comparison infeasibility with possible correlation between literature-reported and own metrics o	☐	☐	
• 7.2 *[If applicable]* reporting of any difference in the kinetic law and relative regression performance between literature-based empirical, semi-mechanistic, or mechanistic models in the same CoU and the current learned relationship o			☐
• 7.3 *[If possible]* if developmental details are available for previous works on the same CoU and their model can be reproduced and trained on own data, reporting of the test set comparison in a fair benchmark setting o	☐	☐	☐
• 7.4 Attach code to the publication or provide a GitHub link with commented functions, used seed and programming language version. Shared codes should follow main good coding practices such as Readme files, inline documentation, function comments, language conventions, and modularity **Δ** (as per data package in regulatory submission)	☐	☐	☐
• 7.5 *[If possible]* attach developmental set to the publication or upload it on a GitHub folder if employed data are sharable, peer-reviewed, and not under any confidentiality restriction. Otherwise, provide a snapshot of how developmental set should look like to run the code, that is, headers (columns name and class-type) and randomly generated first rows **Δ** (as per data package in regulatory submission)	☐	☐	☐

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In this regard, differently from PMx routine, a random or stratified splitting of the entire dataset into development and internal validation sets is mandatory to prove the generalizable capability of the ML-based component. While this is the bare minimum, we strongly suggest embarking on hPMxML efforts only in the presence of external validation sets that are clinically and statistically evaluated as representative of the intended target population or for which covariate shift can be rationalized. In this regard, an upfront comprehensive side-by-side comparison of longitudinal observations or covariate values between developmental and externally collected datasets is mandatory to detect and, in case, quantify this data shift phenomenon. In this latter case, an early interaction with clinical statisticians is strongly suggested for alignments on the setup validity, as well as for strategies to account for it. This is especially relevant in cases where practitioners may want to use real-world data (RWD) sources—electronic health records (EHRs), claims data, patient-reported outcomes, digital health devices—which notoriously have higher dimensionality, more missingness, heterogeneity, and subgroup under-representativeness compared to RCT [75]. In this case, a stringent selection criteria of the baselines and an understanding of the operational differences between clinical practice and trial protocols is fundamental to accomplish translatability and validity assessments [76]. If these look promising, modelers are called to include and detail confounding mitigation strategies, such as censoring-informed propensity score matching (with information on selected distance metric, pairing method, and ratio), in addition to the data processing and model validation items of Table 2.

As part of model robustness analysis, we encourage checking the stability of the ML-selected features (both presence and ranking) across cross-validation folds and a multi-start approach with different seeds towards the verification of influential outliers and of model robustness, respectively. For an overview of the different measures available to assess this consistency, we refer the reader to [77, 78]. In this regard, we warn practitioners to carefully evaluate clinical interpretability, collection costs, and patient burden of each feature before selecting correlation-aware measures, that is, measures that judge as interchangeable highly-correlated features, over unaware ones. In addition, ablation studies via leave-one-out covariate or model part should be conducted to further disentangle the true impact of the component of interest, phasing out the likelihood of mediation or compensation effects.

In terms of hPMxML-specific diagnostics, we believe error propagation quantification, emerging from the parallel or sequential combination of PMx and AI/ML optimization procedures, plays a crucial role in model assessment. Notably, this quantification might be dependent on the integrative engine put in place, that is, the hPMxML category the model belongs to. For instance, for the first category (ML methods testing covariates against target EBEs), a percentage dropout of key covariates could be applied in the final PMx framework to quantify the uncertainty of the ML-based covariate selection. For the second group (regression/classification ML methods including PMx-derived covariates), samples from the conditional distribution could be used to create plausible PMx-derived metrics at the individual level, and error propagation could be computed via bootstrapped training sets incorporating these covariates without replacement. For models in the third category (neural-ODE models automatically retrieving missing terms), readers can refer to [79].

In terms of hPMxML model validation, we state (i) selection of CoU-relevant performance metrics as a trade-off between model assessment and stakeholder familiarity and (ii) explainability of ML components as regulatory cornerstones. For the first, fairness auditing approaches for ethical considerations should be factored in at this stage and guide the identification of an *ad hoc* measure, particularly if preventive strategies have not been applied during data analysis (data augmentation) or model development, or if test sets exhibit imbalances in clinically-relevant covariates, such as age, sex, race, and co-medications. The literature here provides effective metrics for bias evaluation to ensure accurate predictive performances for underrepresented groups. For the second, significant theoretical work has been done in recent years to unbox AI/ML models and render them more understandable and transparent. The compendium of these efforts is represented by feature importance plots [80], identifying which input variables are most influential in building the input–output relationship; partial dependence plots [81], visualizing how outputs change as a consequence of a controlled input variation; SHAP (SHapley Additive exPlanations) plots [82], assigning importance weight to each input for individual output; LIME (Local Interpretable Model-agnostic Explanations) [83], explaining individual outputs through a local approximation of the model. Interestingly, additional methods are available on top of SHAP to convert the feature ranking into clinically-sounding hazard ratios in survival settings [84] or to plug in direct acyclic graphs among features [85], thereby reading feature importance weights from a causal perspective with no confounding effects or mediations.

While the clinical question should clearly translate into a primary research endpoint, we emphasize the International Council for Harmonization ICH E9 recommendation on the use of secondary analysis for additional model verification. Among the others, we strongly suggest performing subgroup analysis on pre-defined and CoU-meaningful cohorts to unravel patient or disease subtype behavior and identify those on which the ML-based model is most performative. In addition, we advise conducting local sensitivity analysis on any pre-set thresholds or cut-offs within clinically reasonable variation space in order to make hPMxML free from developer biases and to prove model robustness.

Finally, to shift towards a unified approach, we advocate for publishing model codes, whether on shared online platforms (as done via the Drug Disease Model Resources (DDMoRE) repository [86]) or as supporting material within a manuscript, to facilitate re-use, re-run, and (occasionally) debug by other practitioners. Given the stringent confidentiality measures on proprietary data in pharma, we acknowledge the difficulty of data sharing on top of modeling scripts in several cases. However, if feasible, we encourage sharing at least a synthetic data template mimicking the format of the real dataset—particularly in terms of headers and column names/types—to guide the work of Clinical Data Programmers.

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8 | Conclusions

The pharmaceutical sector welcomes opportunities to evaluate novel technological and modeling advancements in the clinical routine, provided that these add-ons are well-shaped, endorsed, and extensively validated. Foreseeing opportunities for hPMxML models, we brainstorm around a long-lasting adoptability of these models in clinical drug development. While we recognize some successful ML and hPMxML efforts in the exemplified clinical oncology literature, we note a lack of standardization in the developmental and reporting procedures and some deficiencies in light of GCP and health authority requirements. Although it is also true that many reviewed papers were primarily exploratory rather than designed for guiding drug development decisions, we believe they served as crucial starting points to critically evaluate what is necessary for real-world usage of these technologies across indications. Therefore, to cope with identified gaps in current works, we present our proposal of a comprehensive checklist for optimal and regulatory-compliant hPMxML model building and validation for clinical drug development overall, beyond the oncology disease therapeutic area per se. Indeed, while oncology has been primarily touched on here—as it catalyzed most of hPMxML efforts so far—similar methodological deficiencies were detected in other diseases, as shown in the Supporting Information (Figure S2). Besides, all non-oncological hPMxML applications we revised (see Table S4) employed classical regression or classification tasks, leveraging a diverse range of established theory and libraries. Some of these approaches, however, are still maturing in the context of time-to-event scenarios typical of cancer modeling, thereby making oncology a more challenging use case for us to brainstorm around methodological best practices.

While resulting from an extensive and objective review of works and related pitfalls, we acknowledge that the checklist is produced by a unique pharma, thereby being possibly prone to some biases, and we therefore call the community to action for improvements. Nevertheless, we believe that this first version of the checklist is already in good shape to ensure methodological rigor, reproducibility, and trust among clinical stakeholders, as well as a higher probability of technical success and approval for hPMxML models across all therapeutic areas. We hope that this checklist can strengthen cross collaboration between pharmacometricians, clinical pharmacologists, AI/ML experts, and regulatory agencies to drive innovation in clinical drug development overall. However, it is important for us to caution the Clinical Pharmacology and PMx community against being swayed by AI/ML per se. Clinical modelers should adhere to and build models upon the principle of parsimony, as exemplified by Occam's Razor. Conventional statistical and PMx methods should always be the starting point and the reference, and only when they fail to produce satisfactory results, and under very precise conditions, that is, enough high-quality data, practitioners may want to explore ML-based alternatives.

Author Contributions

A.F., L.B., O.B., P.B.P., C.J., J.L., and F.M. wrote the manuscript. A.F., L.B., O.B., P.B.P., C.J., J.L., and F.M. designed the research. A.F., L.B., O.B., P.B.P., C.J., J.L., and F.M. performed the research.

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Conflicts of Interest

A.F., P.B.P., C.J., and F.M. are current employees of F. Hoffmann-La Roche Ltd. L.B. and O.B. are current employees of Genentech Inc. J.L. is a current employee of A*STAR Bioinformatics Institute. J.L. was an employee of Genentech Inc. at the time of this work. L.B., O.B., P.B.P., C.J., J.L., and F.M. own stock in Roche.

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